

DJ with the Deluge

Manual for the DJ functions of the Deluge firmware 1.2.1-mastertune v19.0 and newer: Scan, Sync, Nudge and the DJ filter

With v19.0 the Deluge plays along with a track from an external player. It **hears** the track's tempo and key (Scan), **takes over** the tempo with one press (Sync), lets you **push** its beat onto the track's by ear, like a jog wheel (Nudge), and gives the whole song a DJ mixer's **filter** on one knob (DJ filter). This manual shows how to wire it, how each function works and how they fit together in a mix.

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Status.

v19.0 has been tested on the PC and with the whole firmware in an emulator that runs the Deluge's real ARM code, **not yet on a real Deluge**. The screen pictures in this manual are the firmware's own OLED images from that emulator. Points marked verify follow from the firmware's code but have not been tested. Keep a copy of your SD card.

Start quiet.

A loud line input and resonant filter sweeps can get loud quickly on headphones.

1. What the DJ functions do

Function	What it does	Where
Scan v18.4	Hears the tempo (BPM) and the key of what comes into the Deluge's input, with its number on the Camelot wheel.	Song view: Settings → Tuning → Scan
Sync v19.0	One press sets the song's tempo to the BPM heard. Your clips then play at the track's tempo.	TAP TEMPO in the Scan view
Nudge v19.0	Bends the Deluge's tempo for a moment, like a jog wheel bends a record, so that its beat lands on the track's.	While playing: hold ◀▶ pressed, turn TEMPO
DJ filter v19.0	One knob for the whole song: low-pass to the left, off in the middle, high-pass to the right.	Song view, filter section: the upper gold knob

The big picture. Beatmatching has two parts: the **same tempo**, and the beats at the **same moment** (the phase). Sync does the first with one press. The second is done by ear: start the Deluge on the track's first beat of a bar, then nudge until the kick drums land together. The DJ filter then gives you the classic moves for transitions and build-ups. The track comes from any player with a line-level output: a phone, a laptop, a CDJ or a media player.

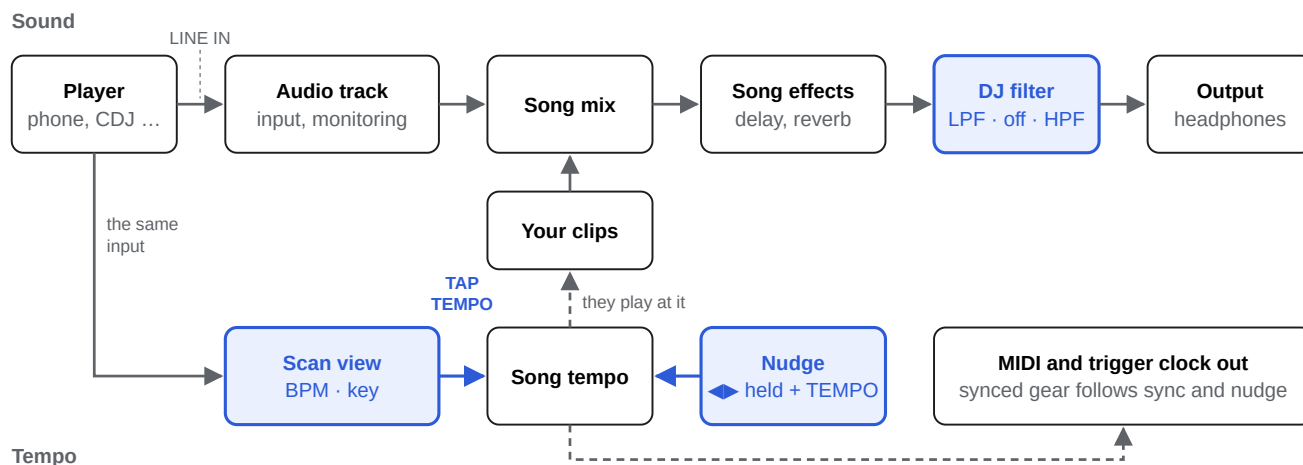


Figure 1. The player through the Deluge (setup A). Blue: the DJ functions. The DJ filter works on the whole song, so on the player too.

2. What you need

- **Firmware v19.0 or newer** (latest: v19.0.1, deluge-1.2.1-mastertune-v19.0.1-l2d-782cf068.bin). Settings → Firmware version shows **V19.0.1-L2D**. **SHIFT** + press **SELECT** opens Settings.
- **A player with a line-level output** and a stereo cable to the Deluge's **LINE IN**. The Deluge has no input gain: you set the level at the player.
- **Headphones or speakers** on the Deluge. The headphones carry the same mix as the main outputs. The Deluge mixes in stereo only with headphones or the right line output plugged in; with only the internal speaker or the left output everything is mono.
- Optional: **a DJ mixer** (setup B, chapter 3.2), and **gear that follows the Deluge's MIDI or trigger clock**: it follows sync and nudge.

3. Setting up

3.1 Setup A: the player through the Deluge

The player comes into the Deluge and out with your song (figure 1). You hear everything from the Deluge, and the DJ filter works on the player as well.

1. **Cable:** the player's output into the Deluge's **LINE IN**.
2. **An audio track for the input:** in Song view press a pad in an **empty row** and keep holding it (a new clip appears), then press **SELECT**: the clip becomes an audio clip on its own audio track. (This works only while the new clip is still empty.)
3. **Its input:** hold **LEARN** and press the audio clip's pad: the menu **AUDIO SOURCE** opens. Turn **SELECT** to **STEREO INPUT (MONITORING)** (7-segment: STER., with the dot) and leave with **BACK**. Only the options with *(monitoring)* pass the input through. The same menu is in the audio clip's own menu (enter the clip, press **SELECT**, *Audio source*).
4. **Test:** start the player. You hear it from the Deluge at once, also while the song is stopped. Set the level at the player (chapter 3.3); the audio track's volume also works on it.

The monitored input passes the audio track's own effects, then the song's effects and the DJ filter. Save the song: the track and its input are kept, so next time only the cable and the player are needed.

3.2 Setup B: with a DJ mixer

The Deluge is one deck of the mixer, the player another. The Deluge only *listens* to the player to read its tempo, so **no** monitoring track: you would hear the player twice. The mixer's cue lets you pre-listen to the Deluge on headphones before you bring it in, which the Deluge alone cannot do. Here the DJ filter works on the Deluge's channel only.

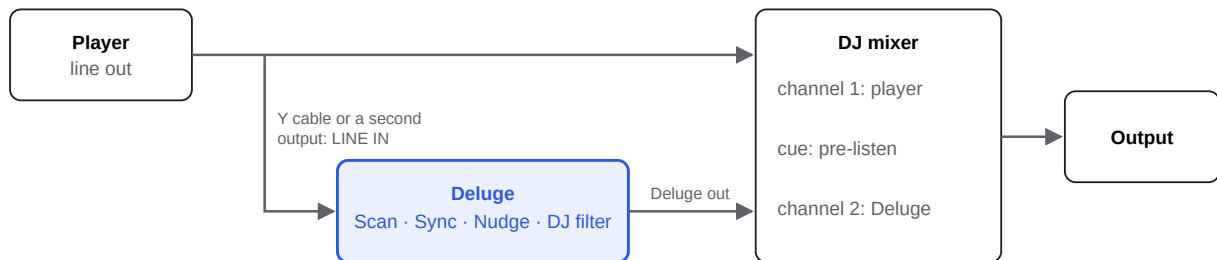


Figure 2. Setup B: the player into the mixer and, split, into the Deluge's LINE IN for Scan. The Deluge's output into its own mixer channel.

For the split, a Y cable at the player's output works (a line output can feed two inputs), or a second output of the player or of the mixer (booth, record or send out).

3.3 The input level

Open the Scan view (chapter 5): its **sidebar** (the audition column, the rightmost) shows the input's level, from –60 dB at the bottom pad to 0 dB at the top, 7.5 dB a pad. Set the player so that the loud parts light **5 to 7 green pads** and the top pad, red, stays dark (it lights above about –4 dB). Too quiet (fewer than 3 pads) makes the tempo hard to read; into the red, the input clips.

4. Quick start

1. Set up as in chapter 3.1. Start the track on the player.
2. In Song view: **[SHIFT]** + press **[SELECT]** → *Tuning* → *Scan: note, BPM, key*. The OLED shows **SCAN LINE**.
3. Press **[SELECT]** until the BPM is big. After about 4 to 5 s of music it shows the tempo, e.g. **128.0 BPM**.
4. If it shows half or double of what you hear, turn **[◀]**: **BPM X2** or **BPM /2**.
5. Press **[TAP TEMPO]**: **SYNC 128.0 BPM**. The song now has the track's tempo.
6. **[BACK]** to Song view. Count along with the track and press **[PLAY]** on a **one** (the first beat of a bar).
7. Listen to the kick drums. The Deluge's comes **after** the track's: hold **[◀]** pressed and turn **[TEMPO]** to the **right**. It comes **before**: to the **left**. Each single detent moves the Deluge about 10 ms.
8. Mix: launch clips, set levels, and for transitions switch the filter section to **DJ** (chapter 8).

5. The Scan view: tempo and key

Open it from Song view: **[SHIFT]** + press **[SELECT]** → *Tuning* → *Scan: note, BPM, key* (7-segment: TUNE, SCAN). The item is there only when you open Settings from Song view. **Leave it** with **[BACK]**, **[SONG]** or **[SCALE]** (lit while Scan is open): back to Song view. The song keeps playing meanwhile, but the pads can't launch clips in the Scan view.



The BPM big. The small line: note and Hz (none here)



The key big, with its Camelot number and the BPM

The firmware's OLED in the emulator, with 128 BPM and A minor set as the analysis would find them.

Control	What it does
Turn SELECT	The input: Auto , Left, Right, Stereo. Auto takes both channels when a jack is plugged in (else the internal microphone). For a player use Auto or Stereo. Each change starts the analysis anew.
Press SELECT	What is big: note, Hz, BPM, key. For DJing the key page is handy: key, Camelot number and BPM at once.
Turn ◀	The BPM shown doubled or halved: BPM X2 , BPM X1 , BPM /2 . The choice stays until you change it or switch the Deluge off, also when you leave Scan and come back.
TAP TEMPO	Sync (chapter 6). SHIFT + TAP TEMPO : the metronome, as everywhere.
Gold knobs	Keep working on the song, the DJ filter included verify .

The title names the input: **SCAN LINE** (a jack in the line input), **SCAN MIC IN**, **SCAN MIC** (the internal microphone), or the channel you chose. A master tune other than 440 Hz follows it, e.g. **SCAN LINE 432**. The 7-segment display shows the big value only: 128.0, AMIN, or ---- for nothing yet.

5.1 Tempo

- Appears after **about 4 to 5 s** of music and is renewed every second: the middle of the last three estimates, each over the last 8 s. After a change of tempo it follows within several seconds.
- Range **60 to 207 BPM**, weighted towards 120. It needs a beat: at least 4 clear hits in 8 s. A beatless intro or breakdown shows **-- BPM**.
- **Half or double**: the music doesn't always say which it is. Drum and bass at 174 often reads as 87, and a slow track can read double. Use to show what you hear.
- In the tests, drum loops read within 0.1 BPM of their tempo.

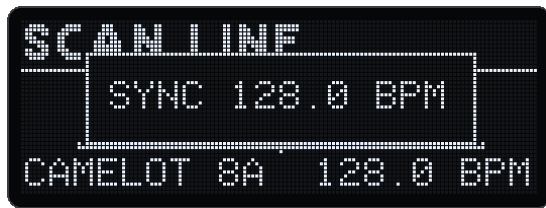
5.2 Key

- Appears after about 4 s of music with notes, follows the music over about 10 s, and goes away after 10 s without notes. Drums alone and a single held note give no key (**--**).
- Major and minor only. On dense music expect some errors to neighbouring keys: the relative key (A minor for C major) or a fifth away.
- **Camelot**: A minor is 8A, C major 8B. Keys that mix well: the **same** number, **one up or down** (8A with 7A or 9A), or the same number with the **other letter** (8A with 8B).

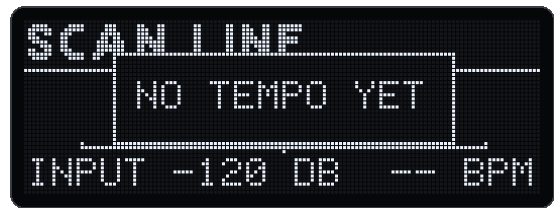
The pads show the tuner's needle (for tuning an instrument); they are not needed for DJing. Scan costs about 2 % of the CPU while it is open and nothing when it is closed.

6. Sync: the song at the track's tempo

In the Scan view, press **TAP TEMPO**. The song's tempo becomes the BPM shown, with $\times 2$ or $\div 2$ applied.



Synced: the song is now at 128 BPM



No tempo heard yet: the tempo stays

- 7-segment: SYNC, or - - - - when there is no tempo yet.
- **While playing** it works at once: the next ticks follow the new tempo, nothing restarts. So you can sync again at any time.
- The song gets the tempo as measured, which can have more decimals than the one shown.
- **Undo:** back in Song view, **BACK** undoes the sync, **SHIFT** + **BACK** redoes it. (In the Scan view **BACK** leaves the view.)
- **EXTERNAL CLOCK** (7-segment EXT): the Deluge follows a MIDI or trigger clock, which sets the tempo. Sync changes nothing then.

Whole numbers. Most electronic tracks have a whole-number tempo. If Scan reads 127.9 and the track is at 128, set 128 exactly: in Song view turn **TEMPO** one detent right and one left. With the community setting *Fine tempo knob* on (the default) the tempo encoder moves in whole BPM, so this lands on the nearest whole number. (On a 7-segment Deluge the first detent only shows the tempo.) An error of 0.1 BPM at 128 moves the beats apart by about 23 ms in 30 s.

7. Nudge: the beats together

While the song plays, in any view, hold the horizontal encoder  **pressed** and turn the **TEMPO** encoder. To the **right** the Deluge moves **ahead** (earlier), to the **left** it moves **back** (later). Each detent shows **SYNC NUDGED** (7-segment NUDGE). In Song view, pressing  first shows the zoom level (e.g. **16TH-NOTES**): that is normal.

How it works

- Each detent makes the tempo **4 % faster** (right) or **slower** (left) for **0.25 s** after it, like a hand on a record. That moves the Deluge by **about 10 ms** at any tempo. Then the tempo is exactly what it was.
- **Keep turning:** the bend stays at 4 %, it doesn't add up. It lasts until 0.25 s after the last detent. So what counts is **how long** you turn: about 40 ms per second of turning, plus 10 ms for the last detent (figure 3).
- For small corrections: single detents with a short pause between them, 10 ms each.



Song view during a nudge right: the bent tempo, 133.33 for 128

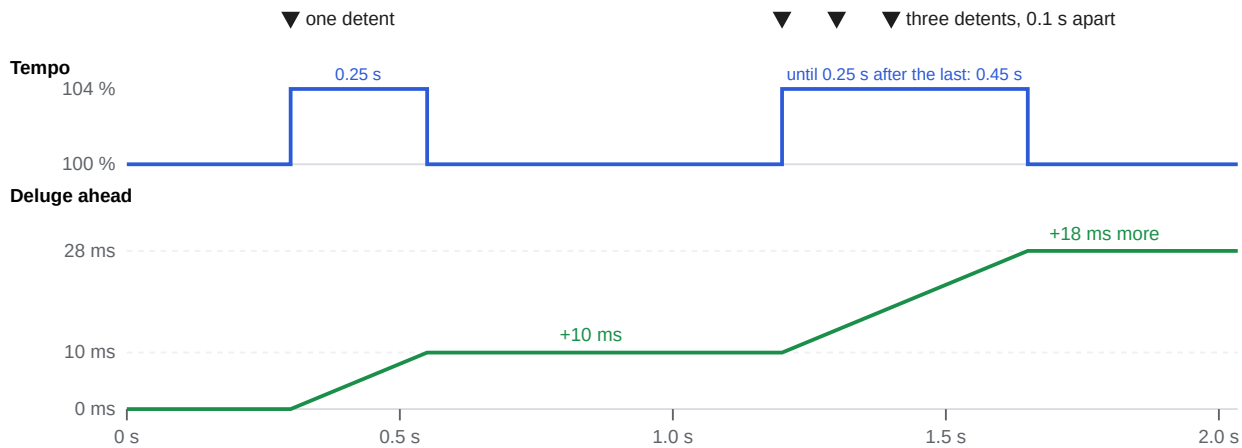


Figure 3. Nudging to the right. A detent bends the tempo about 4 % for 0.25 s: about 10 ms. Three quick detents keep it bent for 0.45 s, not three times as long: about 18 ms.

Situation	Right (clockwise)	Left
Internal clock (the normal case) v19.0	The Deluge moves ahead	The Deluge moves back
SHIFT held too, and the Deluge sends MIDI clock	1.2.1's nudge: one MIDI and trigger clock skipped, the synced gear falls back; the Deluge stays	one clock sent extra, the gear moves ahead; the Deluge stays
External clock (the Deluge follows one)	As in 1.2.1: one clock tick skipped, the Deluge falls back	one tick extra, the Deluge moves ahead

- **Clock outputs follow:** the MIDI clock and the trigger clock output are bent with the Deluge, so gear synced to it moves along.
- **Stopping** (PLAY) ends a bend at once. A song **saved** during a bend saves the tempo without it. If you set the tempo during a bend (tempo encoder, sync), the new tempo stays.
- When stopped, the nudge does nothing: there is no beat to move.
- **Far off?** At 128 BPM a beat lasts 469 ms. For more than about 100 ms, it is quicker to stop and press PLAY again on the next one.

Which way? Listen to the kick drums. Two kicks close together sound like one smeared kick (a "flam"): the beats are 10 to 30 ms apart. If the Deluge's kick is the second of the two, turn right; if it is the first, turn left. When the flam is gone, stop. Check again after a minute.

8. The DJ filter

Switch it on: in Song view (**AFFECT ENTIRE** lit, no clip pad held) the gold knobs work on the whole song. Select the filter section with its mod button (*Cutoff/Resonance*, lit). **Press the upper gold knob:** the filter type goes round **LPF** → **HPF** → **EQ** → **DJ** → LPF. At DJ the popup shows **DJ**. The DJ type exists only for the song: clips and synths go round LPF, HPF and EQ as before.

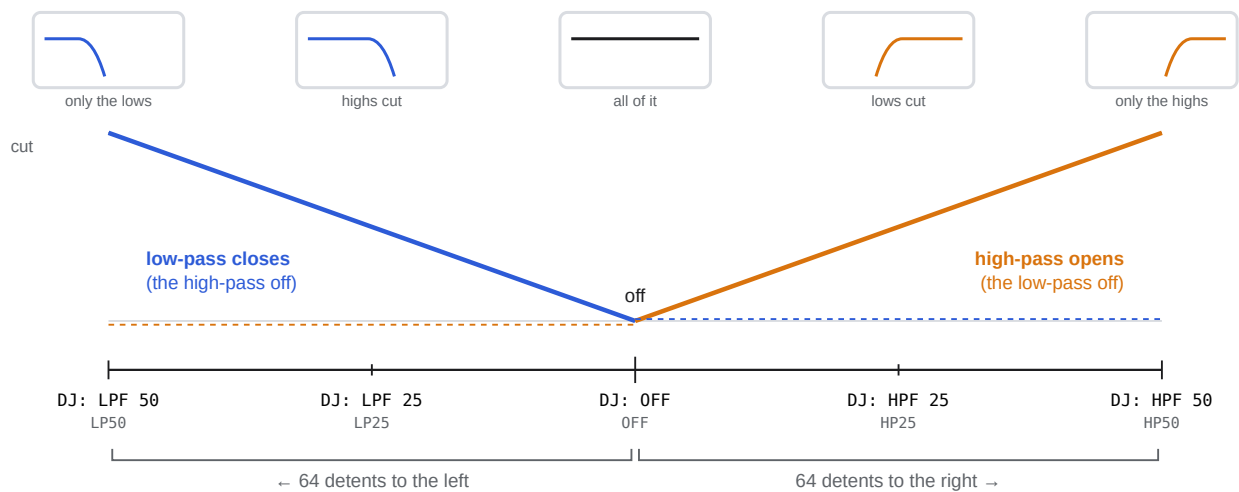
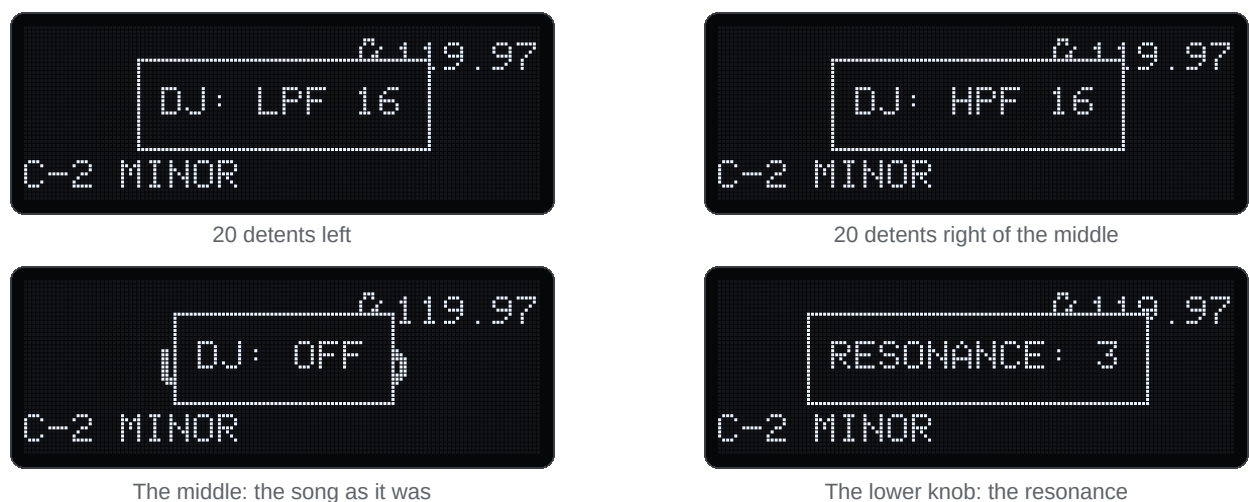


Figure 4. The upper gold knob in DJ: popup on the OLED, below it the 7-segment display. Only one filter is on at a time, each over its whole range.



Control	In DJ
Upper gold knob, turn	Middle DJ: OFF . Left: the low-pass closes, DJ: LPF 1 to DJ: LPF 50 (closed). Right: the high-pass opens, DJ: HPF 1 to DJ: HPF 50 . The numbers use the Deluge's usual 0 to 50 scale. The LEDs at the knob show the position.
Lower gold knob, turn	The resonance of both filters, each from where it is; the popup RESONANCE: 12 (0 to 50) shows the one of the filter that is on.
Upper gold knob, press	On to LPF (the filter type goes round).
Lower gold knob, press	Nothing in DJ. The filters' modes (12 dB, 24 dB ...) are set in LPF and HPF: there the press chooses the mode. DJ uses them as set.

8.1 What it moves

The DJ knob has no filter of its own: it moves the song's own **LPF and HPF cutoff**, exactly as the gold knobs in LPF and HPF would. That is why it sounds like the v18 filters (no stepping, no zipper), and why **saving, undo and**

automation recording work as for those knobs. The song saves its filter type as dj; an older firmware loads LPF there, with the same cutoffs.

- **Set apart before:** if both filters were on (a band-pass set in LPF and HPF, or by automation), the DJ knob goes on from there without a jump. To the right the low-pass opens first until it is off, then the high-pass opens. To the left the high-pass closes first, then the low-pass.
- **Automation** recorded with the DJ knob is automation of the LPF and HPF cutoffs. To delete it, switch to LPF, hold **SHIFT** and press the upper knob: **AUTOMATION DELETED**; the same in HPF for the high-pass. Then press on to DJ.
- **What it filters:** the whole song after its delay and reverb, so their tails are filtered too, and in setup A the player as well. The frequency drone (v12) joins after the song's effects and is not filtered.

Resonance. A little resonance (about 10 to 25 of 50) gives the typical peak of a DJ filter. High values can make the filters sing by themselves, the 24 dB ladder most of all; the firmware doesn't limit it. Start low and raise it while listening.

8.2 Classic moves

- **Build-up:** turn slowly right over 8 or 16 bars: the bass goes away, the mix gets thin and tense. On the drop, back to the middle in one quick turn.
- **Take out, bring in:** close the low-pass slowly (left) to make a part distant and dark, then fade it; bring the next one in from dark to open.
- **Hide the bass during a transition:** with a mixer (setup B), the Deluge's channel a little to the right (high-pass) while the player's bass plays, then swap on a phrase boundary.

9. Mixing with it

9.1 Playing along with a track (setup A)

1. **Prepare a song:** your parts in clips (drums, bass, pads), the audio track for the input (chapter 3.1), the filter section on **DJ** and in the middle. Save it.
2. **Start the track** on the player. Scan, then **TAP TEMPO** (chapters 5, 6). Back to Song view.
3. **Start the Deluge on the one** with a quiet part only (hi-hats, a single clip): mute the rest. Nudge until the kicks or hats land together.
4. **Bring your parts in** by launching or unmuting clips at the start of a phrase (usually every 8 or 16 bars).
5. **Transitions:** the DJ filter on everything, the player's own volume (or the audio track's) for the track alone.
6. **Check the phase every minute** and nudge when it drifts (chapter 9.3).

9.2 With a DJ mixer (setup B)

1. The player on channel 1, the Deluge on channel 2 with its fader down. Sync in Scan as above.
2. Start the Deluge on the one and **pre-listen** on the mixer's cue. Nudge until it sits.
3. Bring it in with the channel fader; the Deluge's DJ filter now works on its channel only, like a channel filter.
4. To go on from the Deluge to the next track, beatmatch the player to the Deluge on the player as usual (its pitch fader), or scan the next track and sync the Deluge to it.

9.3 Staying in time

- Sync gets the tempo to within about 0.1 BPM; that still drifts about 23 ms in 30 s at 128 BPM. Round to the whole number when the track has one (chapter 6), and nudge now and then.

- If the player's tempo changes (a pitch fader moved, a tempo change in the track), sync again: in the Scan view while playing, then nudge.
- A vinyl turntable is never quite steady: expect to nudge more often.

9.4 In key

The key page of the Scan view shows the track's key and Camelot number. Play your melodic parts in that key or a neighbour on the wheel (chapter 5.2): set your clips' scale and root note to it as usual on the Deluge. A one-press "song key from the key heard" is planned for v19.2.

10. Messages

OLED	7-segment	When
DJ	DJ	The upper gold knob pressed: the filter type DJ
DJ: OFF	OFF	The DJ knob in the middle: both filters off
DJ: LPF 1 ... 50	LP1 ... LP50	The DJ knob left of the middle: the low-pass closing
DJ: HPF 1 ... 50	HP1 ... HP50	The DJ knob right of the middle: the high-pass opening
RESONANCE: 12	12	The lower gold knob in DJ
SCAN LINE (title)	the big value, e.g. 128.0	The Scan view open, the input named (7-segment: the menu item is SCAN)
BPM X2 , BPM X1 , BPM /2	the same text	 turned in the Scan view
SYNC 128.0 BPM	SYNC	TAP TEMPO in the Scan view: the song at the tempo heard
NO TEMPO YET	----	TAP TEMPO before a tempo is heard; the tempo stays
EXTERNAL CLOCK	EXT	TAP TEMPO while an external clock sets the tempo; nothing changes
SYNC NUDGED	NUDGE	Each detent of a nudge
16TH-NOTES (or another zoom)	—	 pressed in Song view: the zoom level, before the nudge

11. Troubleshooting

Problem	Likely cause	What to do
No BPM (-- BPM)	No clear beat right now, the input too quiet, or less than 4 s since the start or since the input changed	Wait for a part with drums; check the level (chapter 3.3); for a player use Auto or Stereo
The BPM is half or double	The music allows both readings	Turn  in the Scan view
The BPM jumps between two values	A syncopated or broken rhythm	Sync during a steady part (a four-on-the-floor section), or set the tempo by hand
The beats drift apart after a while	The tempo is off by a fraction, or the player's tempo moved	Round to the whole BPM (chapter 6), sync again, nudge
TAP TEMPO starts the metronome	SHIFT held	Press it without SHIFT, in the Scan view
EXTERNAL CLOCK	The Deluge follows a MIDI or trigger clock	Sync the clock's source, or stop the external clock

The nudge does nothing	The song isn't playing, or  isn't held pressed	Start the song; hold  down while turning TEMPO
The nudge feels weak	A single detent is only 10 ms	Keep turning: about 40 ms per second
Scan is missing in Tuning	Settings was opened from a clip	Go to Song view first
I don't hear the player	No monitoring: the audio track's input without (<i>monitoring</i>), or its volume down	Chapter 3.1, step 3
I hear the player twice	Setup B with a monitoring track as well	Set the audio track's source to one without monitoring, or delete the track
Everything is mono	Only the speaker or the left output plugged in	Headphones or the right line output as well
No DJ after EQ	You are in a clip: DJ is for the song only	Song view, no clip pad held
The filter whistles	Too much resonance	Lower gold knob to the left
The DJ knob starts somewhere unexpected	LPF and HPF were set apart before	Expected: it goes on from there without a jump (chapter 8.1)
An older firmware shows LPF	It doesn't know the DJ type	Expected; the cutoffs are the same

12. Limits and what comes next

- **No cue on the Deluge itself:** the headphones carry the main mix, so pre-listening needs a DJ mixer (setup B).
- **The phase is by ear:** sync sets the tempo, not where the beat falls. There is no automatic beat alignment yet.
- **Tempo reading** needs a beat, 60 to 207 BPM (beyond that with $\times 2$ and $\div 2$), and about 4 s.
- **Key reading:** major and minor only; some errors to neighbouring keys on dense music.
- **One DJ filter** for the whole song mix, not per deck. For one track alone, use that track's own filter.
- **Sync and the new nudge** need the Deluge's own clock. With an external clock the nudge works as in 1.2.1.
- **Not yet tested on a real Deluge** (see the status box on page 1).

Planned (research/DJ-T00L-PLAN.md): **v19.1** the Deluge's own tracks as decks: a computer tool writes each track's tempo into its file, and audio clips then play whole tracks in sync, bar-exact. **v19.2** a crossfader and an isolator EQ (kills for bass, mids and treble) on the Deluge's own controls, and the song's key set from the key heard.

13. Quick reference

Action	Controls	Where, and what you see
Open Settings	SHIFT + press SELECT	Any view
Open Scan	Settings → <i>Tuning</i> → <i>Scan: note, BPM, key</i>	From Song view; SCAN LINE
Leave Scan	BACK , SONG or SCALE	Back to Song view
Scan: input	Turn SELECT	Auto, Left, Right, Stereo
Scan: what is big	Press SELECT	Note, Hz, BPM, key
Scan: ×2 / ÷2	Turn 	BPM X2 , BPM X1 , BPM /2
Sync	TAP TEMPO	In the Scan view; SYNC 128.0 BPM
Undo / redo the sync	BACK / SHIFT + BACK	In Song view
Metronome	SHIFT + TAP TEMPO	Anywhere
Nudge ahead / back	Hold  pressed, turn TEMPO right / left	While playing, any view; SYNC NUDGED ; about 10 ms a detent, 40 ms per second of turning
Nudge only the clock outputs	SHIFT + hold  + turn TEMPO	As 1.2.1 did, when the Deluge sends MIDI clock
Tempo to a whole BPM	Turn TEMPO a detent right, then left	Song view (<i>Fine tempo knob</i> on)
DJ filter on	Filter section; press the upper gold knob until DJ	Song view: LPF → HPF → EQ → DJ
DJ filter	Turn the upper gold knob	Left DJ: LPF 1...50 , middle DJ: OFF , right DJ: HPF 1...50
Resonance	Turn the lower gold knob	RESONANCE: 0...50 ; keep it low at first
Filter modes	In LPF or HPF: press the lower gold knob	12 dB, 24 dB ...; DJ uses them
Delete filter automation	In LPF or HPF: SHIFT + press the upper gold knob	AUTOMATION DELETED
Audio track for the input	Hold a new clip's pad in an empty row + press SELECT	Song view
Its input, with monitoring	Hold LEARN + press the audio clip's pad	STEREO INPUT (MONITORING)

DJ with the Deluge · firmware 1.2.1-mastertune v19.0 and newer · the repository's README . md (section v19.0) has the technical details and the tests.